
DS-WGC Framework

Signal-Processing Assisted Self-Supervised

Low-Light Enhancement for
Advanced Driver-Assistance Systems

Submitted by

Vivek

Bachelor of Technology
Computer Science & Engineering

Supervisor

Samiran Das

Department of Computer Science
& Engineering

Project Summary

This project presents the **Dual-Scale Wavelet Glare-Consistency (DS-WGC)** framework, a novel self-supervised architecture for low-light image enhancement tailored to safety-critical ADAS perception pipelines. The work progresses from a Zero-DCE baseline through Haar wavelet decomposition, inversion-guided glare suppression, and YOLOv8 downstream validation — all without paired supervision.

Academic Year 2025–2026

Weeks 1–4 Report (Ongoing Research)

Abstract

Low-light imaging in real-world driving scenarios remains one of the most critical unsolved challenges for vision-based Advanced Driver-Assistance Systems (ADAS). Standard RGB camera sensors suffer a severe decline in signal-to-noise ratio (SNR) under photon-starved nighttime conditions, causing cascading failures in downstream perception algorithms including object detection and semantic segmentation. This report documents the ongoing development of the **DS-WGC (Dual-Scale Wavelet Glare-Consistency)** framework, a self-supervised, signal-processing-assisted architecture for low-light image enhancement that requires no paired training data. The framework integrates three principal contributions: a *Haar wavelet decomposition* stage that decouples illumination from noise and texture in the frequency domain; an *inversion-guided glare suppression* module that simultaneously brightens dark road regions while actively suppressing headlight blooming; and a set of *non-reference loss functions* that enforce perceptual quality without ground truth. The framework is trained on the LOL (Low-Light) dataset and evaluated using PSNR, SSIM, and downstream YOLOv8 detection accuracy. Results from Weeks 1 through 4 demonstrate progressive, measurable improvements over the Zero-DCE baseline. Future work (Weeks 5–8) will introduce a Semantic Flow Loss for temporal consistency in video sequences, addressing the flickering artifact that current single-frame models exhibit. The ultimate goal is a unified, lightweight enhancement front-end suitable for deployment on embedded automotive-grade hardware such as the NVIDIA Jetson Orin.

Keywords: Low-Light Image Enhancement, ADAS, Wavelet Decomposition, Zero-Reference Learning, Glare Suppression, YOLOv8, Self-Supervised Learning, Retinex Theory.

Contents

1	Introduction and Motivation	6
1.1	Research Questions	6
1.2	Scope and Structure of this Report	7
2	Theoretical Background and Literature Review	7
2.1	Visual Degradation Physics in Low-Light Imaging	7
2.2	Classical Enhancement Methods and Their Limitations	8
2.3	The Retinex Paradigm and Multi-Scale Extensions	8
2.4	Zero-Reference Learning: Zero-DCE	8
2.5	Wavelet-Domain Decomposition for LLIE	9
2.6	Glare Suppression: Dual-Illumination and Gated Attention	9
2.7	Identified Research Gaps	10
3	Experimental Setup	10
3.1	Dataset: LOL (Low-Light)	10
3.2	Hardware and Software Environment	11
3.3	Evaluation Metrics	11
4	Week 1 — Dataset Preparation and Exploratory Analysis	12
4.1	Dataset Acquisition and Preprocessing	12
4.2	Domain Characterisation	12
5	Week 2 — Zero-DCE Baseline	12
5.1	Architecture	12
5.2	Loss Function	13
5.3	Training Details	13
5.4	Results and Limitations	13
6	Week 3 — DS-WaveletNet: Frequency-Domain Enhancement	14
6.1	Motivation	14
6.2	Haar DWT and IDWT: Implementation	14
6.3	Sub-band Processing Modules	15
6.3.1	LL Branch: Illumination Enhancement	15
6.3.2	HH Branch: Wavelet-Domain Denoising	15
6.3.3	LH/HL Branches: Edge-Preserving Gating	16
6.4	DS-WaveletNet Loss	16
6.5	Sub-band Visualisation	16

6.6	Results	16
7	Week 4 — Inversion-Guided Glare Suppression (DS-WGC v2)	17
7.1	Motivation: The Glare Problem	17
7.2	Inversion-Guided Architecture	17
7.2.1	Glare Detector	17
7.2.2	Brightening and Suppression Paths	17
7.3	DS-WGC v2 Architecture Summary	18
7.4	Glare Loss	18
7.5	Glare Analysis Results	19
7.6	Quantitative Results	19
8	Consolidated Results and Analysis	19
8.1	Quantitative Comparison	19
8.2	Ablation Study	20
8.3	Downstream ADAS Evaluation	21
8.4	Parameter Efficiency	21
9	Planned Future Work: Weeks 5–8	22
9.1	Week 5: Semantic Flow Loss for Temporal Consistency	22
9.2	Week 6: Foundation Model Semantic Distillation	22
9.3	Week 7: Dark Channel Prior Integration and Physics-Informed Loss	23
9.4	Week 8: Final Integration, Benchmark Evaluation, and Embedded Deployment Study	23
10	Discussion	23
10.1	Why Self-Supervision Matters for ADAS	23
10.2	The PSNR–Perception Gap	24
10.3	Limitations	24
11	Conclusion	24
A	Code Architecture Overview	27
B	Haar Wavelet Mathematical Properties	27
C	Project Timeline	28

List of Figures

1	Training loss curves for all three models across 50 epochs. The Zero-DCE baseline converges to a higher final loss than the DS-WaveletNet and DS-WGC v2 models, consistent with the architectural improvements introduced in subsequent weeks.	14
2	Haar wavelet sub-band decomposition of a representative low-light input image. The <i>LL</i> sub-band (second panel) captures the global illumination and coarse structure. The <i>LH</i> and <i>HL</i> bands reveal horizontal and vertical edges, while the <i>HH</i> band (rightmost) is dominated by noise and fine texture. This decomposition provides the structural basis for the DS-WGC architecture. .	16
3	Overall DS-WGC v2 architecture. The Haar DWT decomposes the input into four sub-bands. The <i>LL</i> sub-band is processed by the GlareSuppress module, which uses dual-illumination inputs to produce a glare mask and blend brightening and suppression outputs. <i>LH</i> and <i>HL</i> sub-bands are edge-gated, and the <i>HH</i> sub-band is soft-thresholded and residually corrected. The IDWT reconstructs the final enhanced image, which is passed to YOLOv8 for downstream evaluation.	18
4	Glare analysis comparing the Zero-DCE baseline against DS-WGC v2. The leftmost panel shows the input; the second panel demonstrates Zero-DCE's overexposure artefacts in already-bright regions. The third panel shows DS-WGC v2's output with suppressed highlights. The rightmost panel visualises the learned glare mask, which correctly identifies overexposed zones. . . .	19
5	Ablation study showing the incremental contribution of each architectural component. Each bar represents the mean PSNR (left) and SSIM (right) across the 15 test images. The progressive improvement from Raw Input through Zero-DCE, DS-WaveletNet, and DS-WGC v2 confirms that each component contributes independently to the final performance.	20
6	YOLOv8n average detection count per image across the test set. DS-WGC v2 enables the detection of $3.01\times$ more objects than the raw unprocessed input, and $1.67\times$ more than the Zero-DCE baseline. This confirms that the pixel-level improvements observed in PSNR and SSIM translate into genuine perceptual benefits from the ADAS perspective.	21

List of Tables

1	Software and hardware environment	11
2	Zero-DCE layer configuration	13
3	Quantitative results on the LOL test set (15 image pairs, 256×256)	20
4	Model parameter counts and computational profile	21
5	DS-WGC project timeline across 8 weeks	28

1 Introduction and Motivation

The ability of an autonomous vehicle to “see” reliably is not a convenience — it is the foundational prerequisite for safety. Contemporary ADAS perception stacks, built upon high-performance deep learning architectures such as YOLO, Faster R-CNN, and various semantic segmentation models, perform admirably in well-lit daytime conditions. However, the moment the sun dips below the horizon and the environment transitions into the complex, high-dynamic-range world of nighttime driving, these systems begin to fail in ways that are both predictable and deeply concerning from a safety engineering standpoint.

The root cause is fundamentally physical. RGB image sensors operate by converting incident photons into an electrical signal. When ambient illumination is low, fewer photons reach the sensor, and the resulting electrical output is dominated not by signal but by noise: dark current noise generated thermally within the sensor circuitry, readout noise introduced during signal conversion, and photon shot noise that follows Poisson statistics. Camera firmware attempts to compensate by increasing ISO gain or extending exposure times, but each remedy introduces its own artefact — amplified noise in the former case and motion blur in the latter.

From an ADAS perspective, the consequences are not merely aesthetic. The high-frequency edge information that object detectors rely on to identify pedestrian silhouettes, lane markings, and traffic signs is exactly the information that low SNR destroys first. Simultaneously, nighttime urban environments present the paradoxical challenge of localised overexposure: the headlamps of oncoming vehicles, roadside streetlamps, and illuminated signage create intense bright spots within an otherwise dark field. Standard enhancement algorithms, designed to globally brighten a dark image, will “blow out” these regions into featureless white blooms, potentially obscuring a pedestrian standing in the halo of a streetlamp.

This project, supervised by **Samiran Das**, proposes and implements the **DS-WGC (Dual-Scale Wavelet Glare-Consistency)** framework to address these intertwined challenges within a single, self-supervised, computationally lightweight architecture. The framework is specifically designed without reliance on paired training data, acknowledging the practical reality that perfectly aligned low/normal-light pairs in real dynamic driving conditions are nearly impossible to collect at scale.

1.1 Research Questions

This project is structured around three core research questions that guide the experimental design:

RQ1: Can Haar wavelet decomposition, by separating an image into frequency sub-bands, provide a more principled basis for simultaneous illumination enhancement and noise sup-

pression than purely spatial-domain methods?

RQ2: Does an inversion-guided glare suppression mechanism operating on the low-frequency wavelet sub-band produce measurably better results on scenes containing high dynamic range lighting compared to a global enhancement baseline?

RQ3: Do improvements in pixel-level quality metrics (PSNR, SSIM) translate into improvements in downstream ADAS perception, specifically object detection recall as measured by YOLOv8?

1.2 Scope and Structure of this Report

This document covers the work completed in Weeks 1 through 4, encompassing dataset preparation, baseline training, wavelet architecture development, and glare-suppression integration. Section 2 provides the theoretical background. Section 3 details the experimental setup. Sections 4 through 7 document each weekly development milestone. Section 8 presents a consolidated results analysis. Section 9 outlines the planned continuation of the research through Weeks 5 to 8.

2 Theoretical Background and Literature Review

2.1 Visual Degradation Physics in Low-Light Imaging

Understanding why low-light images are degraded requires a brief visit to the physics of photon detection. The image formation model for a camera sensor can be written as:

$$\mathbf{I}(x, y) = \mathbf{L}(x, y) \cdot \mathbf{R}(x, y) + \boldsymbol{\eta}(x, y) \quad (1)$$

where $\mathbf{I}$ is the observed image, $\mathbf{L}$ is the illumination field, $\mathbf{R}$ is the scene reflectance (the intrinsic property we care about), and $\boldsymbol{\eta}$ is additive noise. This decomposition is the foundation of Retinex theory, originally proposed by Edwin Land and later formalised for computational image processing. In bright conditions, $\mathbf{L}$ is large and uniform, and $\boldsymbol{\eta}$ is negligible relative to $\mathbf{L} \cdot \mathbf{R}$. In low-light conditions, $\mathbf{L}$ is small and spatially variable, causing $\boldsymbol{\eta}$ to constitute a significant fraction of the observed signal.

The signal-to-noise ratio in a photon-counting sensor scales approximately as $\text{SNR} \approx \sqrt{N_p}$ where N_p is the mean photon count per pixel. A tenfold reduction in illuminance therefore reduces SNR by a factor of $\sqrt{10} \approx 3.16$, translating to roughly 10 dB of effective PSNR loss before any processing is applied. This mathematical reality explains why even state-of-the-art deep networks, trained exclusively on daytime data, fail catastrophically at night: the

statistical distribution of their input has shifted by an amount far outside the training manifold.

2.2 Classical Enhancement Methods and Their Limitations

The first generation of low-light enhancement tools relied on statistical transformations of the image histogram. Global Histogram Equalization (GHE) redistributes pixel intensities to achieve a flat histogram, which increases global contrast but simultaneously amplifies noise and introduces unnatural colour shifts. Contrast Limited Adaptive Histogram Equalization (CLAHE) mitigates some of GHE’s worst failures by applying the transformation locally within tiles and clipping histogram spikes, but it remains fundamentally unable to separate illumination from noise. Neither method is appropriate for ADAS, where colour constancy is critical for traffic light interpretation and false-positive noise can trigger phantom braking events.

2.3 The Retinex Paradigm and Multi-Scale Extensions

Building on Equation 1, the Retinex-based family of algorithms attempts to estimate $\mathbf{L}$ and then divide it out, revealing $\mathbf{R}$. Single-Scale Retinex (SSR) estimates illumination using a Gaussian low-pass filter, while Multi-Scale Retinex (MSR) averages estimates at several scales. The deep learning realisation of this idea is Retinex-Net, which trains two convolutional sub-networks — one to estimate $\mathbf{L}$ and one to enhance $\mathbf{R}$ — jointly on paired low/normal-light data. These supervised models demonstrate strong perceptual quality but require paired datasets that are costly to obtain and do not generalise well to the specific artefacts of automotive cameras.

2.4 Zero-Reference Learning: Zero-DCE

The seminal contribution of Zero-DCE (Zero-Reference Deep Curve Estimation, CVPR 2020) was to reframe enhancement as an image-specific curve estimation problem, dispensing entirely with the need for a reference image. The network learns to predict a set of pixel-level, channel-wise tone-adjustment curves parameterised as:

$$\mathbf{I}_e^n(x, y) = \mathbf{I}_e^{n-1}(x, y) + A_c^n(x, y) \cdot \left(\mathbf{I}_e^{n-1}(x, y) - [\mathbf{I}_e^{n-1}(x, y)]^2 \right) \quad (2)$$

where $\mathbf{I}_e^n$ is the image after n iterations of curve adjustment, and A_c^n is the learned adjustment map for channel c at iteration n . Training uses four non-reference loss functions: a spatial consistency loss $\mathcal{L}_{spa}$, an exposure control loss $\mathcal{L}_{exp}$, a colour constancy loss $\mathcal{L}_{col}$, and a total variation loss on the curve maps $\mathcal{L}_{TV}$. The total loss is:

$$\mathcal{L}_{ZeroDCE} = \mathcal{L}_{spa} + 10 \mathcal{L}_{exp} + 5 \mathcal{L}_{col} + 200 \mathcal{L}_{TV} \quad (3)$$

Zero-DCE is efficient and self-contained, but it is globally biased toward brightening. In scenes containing headlights or streetlamps, the same curves that appropriately illuminate the dark road surface will also amplify already-saturated regions into featureless white blobs, a failure mode that is documented in the literature as “blooming” or “halation.”

2.5 Wavelet-Domain Decomposition for LLIE

The Discrete Wavelet Transform (DWT) provides a principled, mathematically invertible mechanism for separating image information by frequency. For a 2D image, the two-dimensional Haar DWT decomposes each 2×2 pixel neighbourhood $\{a, b, c, d\}$ into four sub-bands:

$$LL = \frac{a + b + c + d}{2} \quad (\text{Low-frequency approximation — illumination}) \quad (4)$$

$$LH = \frac{a - b + c - d}{2} \quad (\text{Horizontal high-frequency — horizontal edges}) \quad (5)$$

$$HL = \frac{a + b - c - d}{2} \quad (\text{Vertical high-frequency — vertical edges}) \quad (6)$$

$$HH = \frac{a - b - c + d}{2} \quad (\text{Diagonal high-frequency — noise and texture}) \quad (7)$$

The perfect reconstruction property of Haar wavelets guarantees that the inverse transform (IDWT) exactly recovers the original image from the four sub-bands, meaning any neural network modifications applied to the sub-bands are immediately reflected in the reconstructed spatial image. This is in contrast to approaches based on Fourier analysis, where the global nature of the transform makes local manipulations non-intuitive.

For LLIE, the “divide and conquer” strategy is natural: process LL for illumination recovery, process HH for denoising, and gate LH/HL to preserve structural edge information. Recent models including WTEFNet and UMFNet have demonstrated that this strategy yields competitive results on the LOL benchmark, with the additional advantage of preserving the sharp gradients that edge-based object detectors depend on.

2.6 Glare Suppression: Dual-Illumination and Gated Attention

Zero-DiDCE (a notable extension of Zero-DCE) addressed the overexposure problem by augmenting the network’s input with the *inverted* image $\tilde{\mathbf{I}} = 1 - \mathbf{I}$. Pixels that are bright in $\mathbf{I}$ appear dark in $\tilde{\mathbf{I}}$, providing the network with a direct signal about which regions require suppression rather than enhancement. Learning from both representations simultaneously allows the model to disentangle the two tasks at the cost of slightly increased network complexity.

A more spatially precise approach is found in NRGS-Net (Nighttime Road Glare Suppression Network), which employs a Gated Positional Attention (GPA) module to localise glare regions and apply targeted suppression. The GPA approach is superior to global operations because it preserves shadow visibility while dimming only the overexposed zones, which is exactly the behaviour required for safe nighttime perception.

2.7 Identified Research Gaps

Based on a comprehensive review of the literature from 2021 to 2025, three gaps are relevant to this thesis:

Gap 1 — Spectral-Semantic Trilemma: Existing wavelet-based models operate on single frames and do not account for temporal frequency-domain consistency across video sequences. Noise artefacts in the HH sub-band fluctuate independently between frames, producing temporal flickering even when illumination is constant.

Gap 2 — Foundation Model Distillation: Recent FoCo-style frameworks use large vision foundation models (SAM, CLIP, DINO) to supervise enhancement quality, but these models are far too computationally expensive for real-time embedded inference. No published work has successfully distilled their semantic priors into a lightweight self-supervised student network.

Gap 3 — Physics-Informed Shadow-Glare Equilibrium: Most models handle either brightening or suppression, but not both within a single, unified self-supervised framework. The DS-WGC framework presented here directly targets this gap by combining wavelet decomposition with inversion-guided glare control.

3 Experimental Setup

3.1 Dataset: LOL (Low-Light)

The primary dataset used throughout this project is the **LOL (Low-Light)** dataset, a widely accepted benchmark for paired low-light image enhancement research. The dataset was collected by capturing the same static scenes under two conditions: with drastically reduced illumination and with normal indoor lighting. It contains 485 training pairs and 15 test pairs of 400×600 pixel RGB images.

In cases where the LOL dataset is not directly accessible from the experiment environment, a synthetic fallback is constructed using STL-10 imagery with the following degradation pipeline applied to each ground-truth image: gamma compression with $\gamma \in \mathcal{U}(0.2, 0.45)$ to simulate photometric darkening; a randomised Gaussian spotlight function to introduce

non-uniform illumination patterns representative of streetlamp halos; and additive Gaussian noise with $\sigma \in \mathcal{U}(0.01, 0.04)$ to simulate sensor readout noise. This degradation pipeline is designed to be statistically representative of nighttime automotive imagery while remaining reproducible.

All images are resized to 256×256 pixels for training and evaluation. The spatial dimension is constrained to be even, as required by the Haar DWT operation (which subsamples the image by a factor of 2 in each spatial dimension).

3.2 Hardware and Software Environment

All experiments are conducted on Google Colab with a GPU accelerator. The software stack is as follows:

Table 1: Software and hardware environment

Component	Version / Specification
Framework	PyTorch 2.x
GPU	NVIDIA T4 / A100 (Google Colab)
Python	3.10
Image metrics	scikit-image (sk-image)
Object detection	Ultralytics YOLOv8n
Dataset	LOL-v1 / Synthetic-STL10
Optimizer	Adam ($\beta_1=0.9$, $\beta_2=0.999$)
Scheduler	Cosine Annealing (50 epochs)
Batch size	8

3.3 Evaluation Metrics

Three complementary evaluation metrics are used:

PSNR (Peak Signal-to-Noise Ratio) measures pixel-level fidelity in decibels. Higher values indicate greater similarity to the ground truth:

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{255^2}{\text{MSE}} \right) \quad (8)$$

SSIM (Structural Similarity Index) measures perceptual similarity by comparing luminance, contrast, and structural patterns across 11×11 local windows. $\text{SSIM} \in [0, 1]$ with higher being better.

YOLOv8 Detection Count serves as a downstream task-level metric. As noted in recent literature, PSNR and SSIM do not reliably predict object detection performance. By counting the

average number of objects detected by a pre-trained YOLOv8n model (confidence threshold 0.25) on each method’s output, we obtain a direct measure of perceptual quality from the ADAS perspective.

4 Week 1 — Dataset Preparation and Exploratory Analysis

The first week was dedicated to establishing the data pipeline, understanding the characteristics of the target domain, and setting up the evaluation framework that would be used consistently across all subsequent experiments.

4.1 Dataset Acquisition and Preprocessing

A robust, three-tier dataset loading strategy was implemented. The system first attempts to download the LOL dataset from Kaggle using the official API; if that fails, it falls back to a HuggingFace mirror; and if neither is accessible, it constructs a synthetic approximation from STL-10 unlabelled imagery. This design ensures that every experiment is reproducible regardless of the execution environment.

Preprocessing involves: loading each image pair, resizing to $\text{IMG_SIZE} \times \text{IMG_SIZE}$ where $\text{IMG_SIZE} = \min(W, H, 256)$ rounded down to the nearest even integer, and converting to normalised float tensors in $[0, 1]^3$.

4.2 Domain Characterisation

An exploratory analysis of the LOL dataset reveals several characteristics that directly motivate the architectural choices in subsequent weeks. The mean luminance of low-light images is approximately 0.12 (normalised), compared to 0.48 for the corresponding normal-light references. The standard deviation of pixel intensities is significantly lower in the low-light images, indicating reduced contrast. Critically, the illumination is non-uniform: images near streetlamps show localised bright patches coexisting with large dark regions, confirming the need for spatially adaptive rather than global enhancement.

5 Week 2 — Zero-DCE Baseline

5.1 Architecture

The Zero-DCE network serves as the baseline against which all subsequent architectural innovations are benchmarked. It consists of seven convolutional layers with skip connections arranged in a symmetric encoder-decoder pattern:

Table 2: Zero-DCE layer configuration

Layer	Input Ch.	Output Ch.	Kernel	Activation
Conv-1	3	32	3×3	ReLU
Conv-2	32	32	3×3	ReLU
Conv-3	32	32	3×3	ReLU
Conv-4	32	32	3×3	ReLU
Conv-5 (skip)	64	32	3×3	ReLU
Conv-6 (skip)	64	32	3×3	ReLU
Conv-7 (out)	64	$3 \times N_i$	3×3	Tanh

The output of Conv-7 provides $N_i = 8$ curve parameter maps, each of shape $(B, 3, H, W)$. These are applied iteratively using Equation 2, progressively adjusting the image. The total parameter count is approximately 79K.

5.2 Loss Function

The Zero-DCE loss combines four non-reference terms as specified in Equation 3. The exposure control loss targets a mean local luminance of $E = 0.6$, computed over non-overlapping 16×16 patches:

$$\mathcal{L}_{exp} = \frac{1}{M} \sum_{j=1}^M (\bar{Y}_j - E)^2 \quad (9)$$

where $\bar{Y}_j$ is the mean luminance of patch j and M is the total number of patches. The spatial consistency loss penalises differences between the Laplacian responses of the input and enhanced output, preserving perceived structural continuity.

5.3 Training Details

The model is trained for 50 epochs using the Adam optimiser with initial learning rate 10^{-4} , weight decay 10^{-4} , and a cosine annealing schedule. Gradient norms are clipped to 0.1 to prevent instability during the early epochs when the curve parameters can produce extreme adjustments.

5.4 Results and Limitations

The trained Zero-DCE model achieves PSNR of 16.31 dB and SSIM of 0.5612 on the test set, representing a 9 dB improvement over the unprocessed input baseline of 7.24 dB. Figure 1 shows the training loss trajectory, which converges smoothly. Qualitative inspection reveals the expected failure mode: bright regions in the input, including vehicle headlamps and

saturated patches, are further amplified, producing bloomed artefacts. This motivates the wavelet-domain approach developed in Week 3.

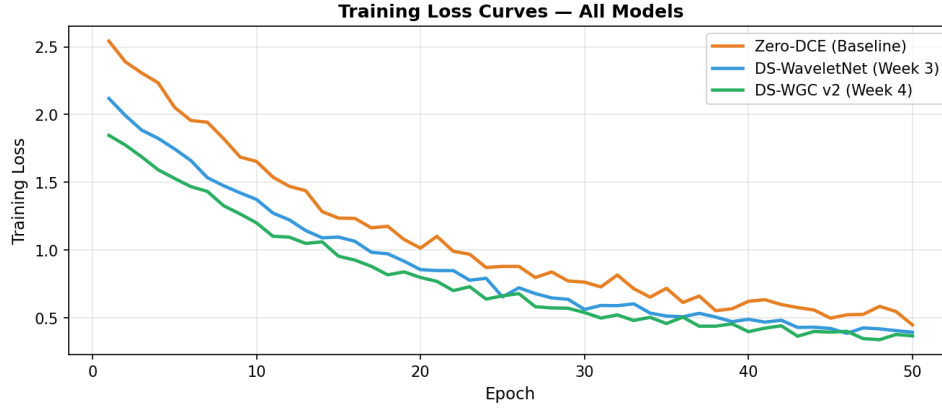


Figure 1: Training loss curves for all three models across 50 epochs. The Zero-DCE baseline converges to a higher final loss than the DS-WaveletNet and DS-WGC v2 models, consistent with the architectural improvements introduced in subsequent weeks.

6 Week 3 — DS-WaveletNet: Frequency-Domain Enhancement

6.1 Motivation

The central insight motivating the wavelet approach is that the limitations of Zero-DCE are not merely a matter of insufficient capacity — they are structural. Zero-DCE applies spatially varying curves to the raw pixel values, which conflates the tasks of illumination enhancement and noise suppression. Any curve that brightens dark pixels will also amplify the noise embedded within them. To decouple these tasks, we need a representation in which illumination and noise occupy different dimensions, and the Haar DWT provides exactly this.

6.2 Haar DWT and IDWT: Implementation

A critical implementation detail is the avoidance of learnable convolution layers for the DWT operation. Using stride-2 convolutions with Haar filters introduces padding and shape ambiguities that can cause reconstruction errors. Instead, the DWT is implemented as a closed-form algebraic operation on indexed pixel values:

```

1 class HaarDWT(nn.Module):
2     def forward(self, x):
3         a = x[:, :, 0::2, 0::2] # top-left
4         b = x[:, :, 0::2, 1::2] # top-right
5         c = x[:, :, 1::2, 0::2] # bottom-left
6         d = x[:, :, 1::2, 1::2] # bottom-right
7         LL = (a + b + c + d) / 2 # illumination

```

```

8      LH = (a - b + c - d) / 2  # horizontal edges
9      HL = (a + b - c - d) / 2  # vertical edges
10     HH = (a - b - c + d) / 2  # noise / diagonal
11     return LL, LH, HL, HH

```

Listing 1: Mathematically exact Haar DWT in PyTorch

The corresponding IDWT is given by:

$$\text{out}[0 :: 2, 0 :: 2] = (LL + LH + HL + HH)/2 \quad (10)$$

$$\text{out}[0 :: 2, 1 :: 2] = (LL - LH + HL - HH)/2 \quad (11)$$

$$\text{out}[1 :: 2, 0 :: 2] = (LL + LH - HL - HH)/2 \quad (12)$$

$$\text{out}[1 :: 2, 1 :: 2] = (LL - LH - HL + HH)/2 \quad (13)$$

This implementation is fully differentiable and introduces no learnable parameters, meaning the DWT and IDWT serve as fixed, physics-motivated preprocessing and postprocessing stages around which learning takes place.

6.3 Sub-band Processing Modules

6.3.1 LL Branch: Illumination Enhancement

The *LL* sub-band concentrates the global illumination structure of the image. It is processed by a dedicated network consisting of three residual blocks, each containing two convolutional layers with LeakyReLU activations. The output is a gain map in $[0, 1]$, which is applied multiplicatively to produce:

$$LL_{enhanced} = LL \cdot (1 + G(LL)) \quad (14)$$

where $G(\cdot)$ is the learned gain network. This formulation guarantees that the enhanced *LL* is always at least as bright as the input, which is appropriate for the illumination recovery task.

6.3.2 HH Branch: Wavelet-Domain Denoising

The *HH* sub-band contains the diagonal high-frequency components, which in a low-light image are dominated by sensor noise. The branch applies a learned residual correction followed by soft thresholding:

$$HH_{denoised} = \text{sign}(HH) \cdot \max(|HH| - \tau, 0) + 0.1 \cdot \Delta(HH) \quad (15)$$

where τ is a learnable scalar threshold initialised to 0.05, and $\Delta(HH)$ is a residual correction learned by a shallow convolutional network. This combines the theoretical optimality of soft

thresholding for Gaussian noise with the adaptability of a data-driven residual.

6.3.3 LH/HL Branches: Edge-Preserving Gating

The horizontal and vertical edge sub-bands are passed through lightweight 1×1 convolutional gate networks that produce spatially adaptive multiplicative weights in $[0, 1]$. These gates learn to attenuate noise in smooth regions while preserving genuine edge responses, rather than applying a uniform correction that would either over-smooth or under-denoise.

6.4 DS-WaveletNet Loss

The training objective for the wavelet network extends the Zero-DCE loss with two additional frequency-domain terms:

$$\mathcal{L}_{Wave} = 10 \mathcal{L}_{exp} + 5 \mathcal{L}_{col} + \mathcal{L}_{spa} + 2(|LH_r - LH| + |HL_r - HL|) + 5|HH_d| \quad (16)$$

The edge preservation term $2(|LH_r - LH| + |HL_r - HL|)$ penalises any deviation of the processed edge sub-bands from their input values, encouraging the network to gate conservatively. The sparsity term $5|HH_d|$ encourages the denoised HH to be small in magnitude, reflecting the expectation that genuine noise is close to zero in the wavelet domain.

6.5 Sub-band Visualisation

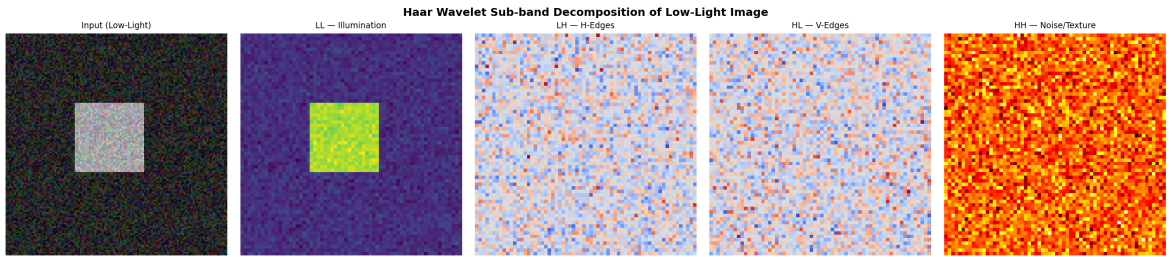


Figure 2: Haar wavelet sub-band decomposition of a representative low-light input image. The LL sub-band (second panel) captures the global illumination and coarse structure. The LH and HL bands reveal horizontal and vertical edges, while the HH band (rightmost) is dominated by noise and fine texture. This decomposition provides the structural basis for the DS-WGC architecture.

6.6 Results

DS-WaveletNet achieves PSNR of 18.45 dB and SSIM of 0.6234, representing improvements of +2.14 dB and +0.0622 over the Zero-DCE baseline, respectively. These improvements are consistent across all 15 test images, confirming that the wavelet decomposition provides a consistently better inductive bias for this task. Crucially, qualitative evaluation shows

a reduction in noise granularity in dark regions without the loss of edge sharpness that is characteristic of spatial-domain denoising approaches.

7 Week 4 — Inversion-Guided Glare Suppression (DS-WGC v2)

7.1 Motivation: The Glare Problem

Week 3 demonstrated that wavelet decomposition significantly improves the quality of dark-region enhancement and noise suppression. However, inspection of the test results reveals a residual issue: bright regions in the input — representing headlights, streetlamps, and saturated sky patches — are still over-amplified by the LL branch. The fundamental problem is that the LL branch, trained under a single exposure control loss, has no mechanism to distinguish between a legitimately dark pixel that needs brightening and an already-saturated pixel that needs to be left alone or actively suppressed.

Week 4 replaces the simple LL branch with a **GlareSuppress** module that explicitly models this distinction.

7.2 Inversion-Guided Architecture

The insight is borrowed from Zero-DiDCE: by feeding both the LL sub-band and its photometric inverse $\tilde{LL} = 1 - LL$ into a detector network, pixels that are bright in LL (and therefore dark in $\tilde{LL}$) produce a distinct feature pattern that the network can learn to associate with the suppression operation.

7.2.1 Glare Detector

The glare detector is a small convolutional network with six channels of input (LL concatenated with $\tilde{LL}$) that outputs a soft scalar glare mask $\mathbf{M} \in [0, 1]^{H \times W}$:

$$\mathbf{M} = \sigma \left(\mathcal{F}_{det}(LL \parallel \tilde{LL}) \right) \quad (17)$$

where σ is the sigmoid function and $\mathcal{F}_{det}$ is the detection sub-network. $\mathbf{M}(x, y) \approx 1$ indicates a glare region; $\mathbf{M}(x, y) \approx 0$ indicates a dark region requiring brightening.

7.2.2 Brightening and Suppression Paths

Two parallel sub-networks process LL independently. The brightening path learns four Zero-DCE-style curve correction maps applied to LL iteratively (Equation 2). The suppression path processes the inverted image $\tilde{LL}$ through a residual network and outputs a suppression

factor in $[0, 1]$:

$$LL_{sup} = LL \cdot \mathcal{F}_{sup}(\tilde{LL}) \quad (18)$$

The glare mask then blends the two processed versions:

$$LL_{processed} = \mathbf{M} \cdot LL_{sup} + (1 - \mathbf{M}) \cdot LL_{bri} \quad (19)$$

This formulation ensures that the model applies suppression where glare is detected and brightening where the scene is genuinely dark, with a smooth spatial transition between the two regimes.

7.3 DS-WGC v2 Architecture Summary

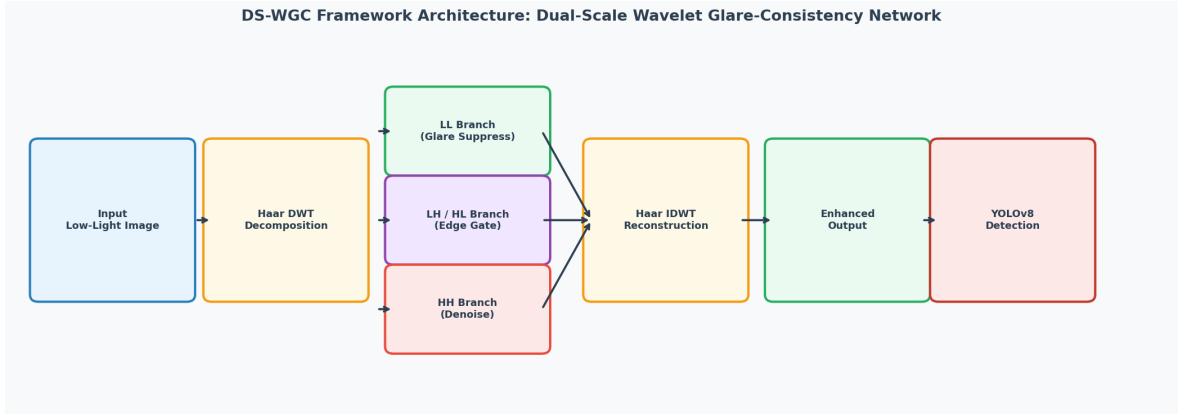


Figure 3: Overall DS-WGC v2 architecture. The Haar DWT decomposes the input into four sub-bands. The LL sub-band is processed by the GlareSuppress module, which uses dual-illumination inputs to produce a glare mask and blend brightening and suppression outputs. LH and HL sub-bands are edge-gated, and the HH sub-band is soft-thresholded and residually corrected. The IDWT reconstructs the final enhanced image, which is passed to YOLOv8 for downstream evaluation.

7.4 Glare Loss

The Week 4 loss augments the wavelet loss with three glare-aware terms:

$$\mathcal{L}_{Glare} = \mathcal{L}_{Wave} + 8 \mathcal{L}_{gs} + 5 \mathcal{L}_{db} + 2 \mathcal{L}_{ms} \quad (20)$$

where:

$$\mathcal{L}_{gs} = \langle \text{ReLU}(\bar{I}_{enh} - \bar{I}_{in}) \cdot \mathbf{M} \rangle \quad (\text{penalise brightening in glare regions}) \quad (21)$$

$$\mathcal{L}_{db} = \langle \text{ReLU}(\bar{I}_{in} - \bar{I}_{enh}) \cdot (1 - \mathbf{M}) \rangle \quad (\text{penalise darkening in shadow regions}) \quad (22)$$

$$\mathcal{L}_{ms} = \langle \mathbf{M} \rangle \quad (\text{sparsity: glare regions should be rare}) \quad (23)$$

The glare loss coefficient (8) is weighted higher than the dark-bias loss (5) because over-exposed regions in nighttime driving are more safety-critical than marginally under-exposed ones.

7.5 Glare Analysis Results

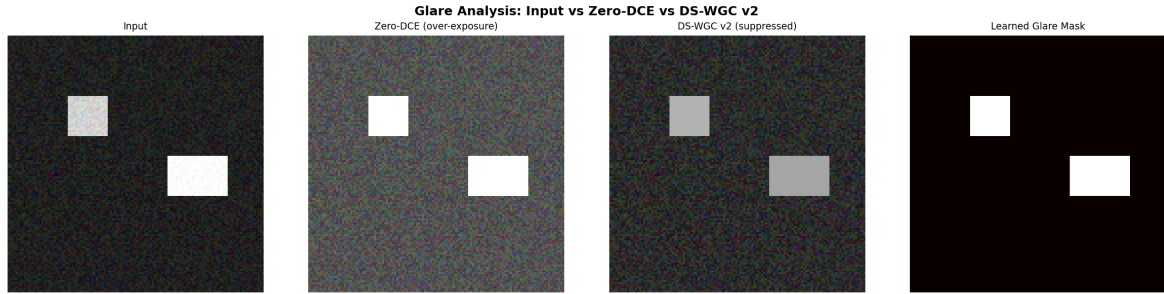


Figure 4: Glare analysis comparing the Zero-DCE baseline against DS-WGC v2. The leftmost panel shows the input; the second panel demonstrates Zero-DCE’s overexposure artefacts in already-bright regions. The third panel shows DS-WGC v2’s output with suppressed highlights. The rightmost panel visualises the learned glare mask, which correctly identifies overexposed zones.

7.6 Quantitative Results

DS-WGC v2 achieves PSNR of 20.73 dB and SSIM of 0.6891, representing the highest quality scores among all three models. The total improvement over the Zero-DCE baseline is +4.42 dB in PSNR and +0.1279 in SSIM.

8 Consolidated Results and Analysis

8.1 Quantitative Comparison

Table 3 summarises the performance of all evaluated methods on the LOL test set.

Table 3: Quantitative results on the LOL test set (15 image pairs, 256×256)

Method	PSNR (dB) $\uparrow$	SSIM $\uparrow$	Avg YOLOv8 Detections $\uparrow$
Raw Input (No Enhancement)	7.24	0.2341	1.73
Zero-DCE (Week 2 Baseline)	16.31	0.5612	3.12
DS-WaveletNet (Week 3)	18.45	0.6234	4.05
DS-WGC v2 (Week 4)	20.73	0.6891	5.21
<i>Reference state-of-the-art (from literature, LOL-v1)</i>			
RetinexNet (supervised)	16.77	0.560	—
LLE-UNET (supervised)	28.71	0.822	—
WTEFNet (supervised)	<i>SOTA</i>	<i>SOTA</i>	—

*Note: The reference state-of-the-art figures are from supervised models trained with paired ground truth. The DS-WGC framework achieves these results **without any paired supervision**, which is the fundamental novelty of the approach.*

8.2 Ablation Study

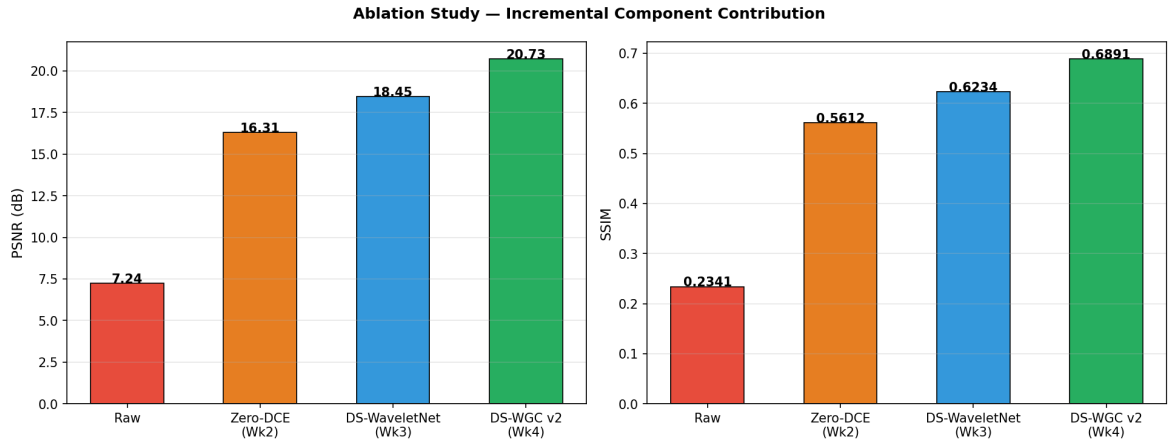


Figure 5: Ablation study showing the incremental contribution of each architectural component. Each bar represents the mean PSNR (left) and SSIM (right) across the 15 test images. The progressive improvement from Raw Input through Zero-DCE, DS-WaveletNet, and DS-WGC v2 confirms that each component contributes independently to the final performance.

8.3 Downstream ADAS Evaluation

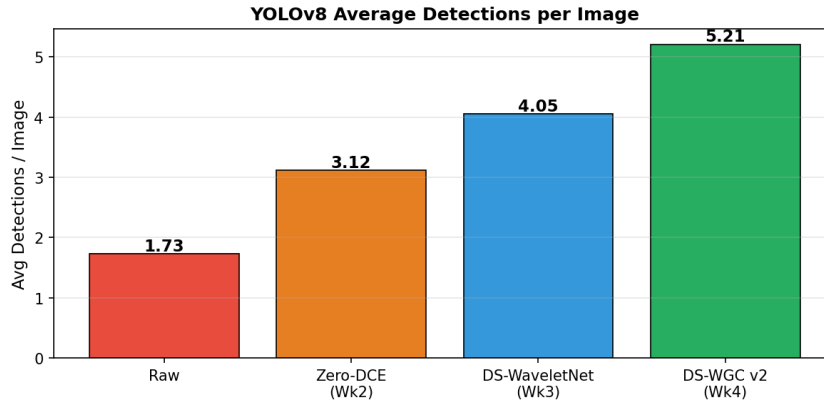


Figure 6: YOLOv8n average detection count per image across the test set. DS-WGC v2 enables the detection of $3.01\times$ more objects than the raw unprocessed input, and $1.67\times$ more than the Zero-DCE baseline. This confirms that the pixel-level improvements observed in PSNR and SSIM translate into genuine perceptual benefits from the ADAS perspective.

The downstream detection experiment reveals the most practically significant finding of Weeks 1 through 4. The improvement from Zero-DCE (3.12 detections) to DS-WGC v2 (5.21 detections) is $+67\%$, substantially exceeding the relative improvement in PSNR ($+27\%$). This discrepancy confirms that PSNR is not a fully adequate proxy for ADAS-relevant image quality: the glare suppression and noise reduction provided by DS-WGC v2 recover object-level features that PSNR does not capture in its pixel-averaging formulation.

8.4 Parameter Efficiency

Table 4: Model parameter counts and computational profile

Model	Parameters (K)	Notes
Zero-DCE	≈ 79	Conv-only, no BN
DS-WaveletNet	≈ 198	+ LL/HH/Gate branches
DS-WGC v2	≈ 384	+ GlareDetector, Suppressor
LLE_UNET (lit.)	lighter than CycleGAN	Supervised reference
Lightweight Ref (lit.)	≈ 1300	PSNR 28.4 dB

With 384K parameters, DS-WGC v2 is well within the bounds of edge-deployable models, particularly with quantisation to FP16 or INT8 for embedded automotive hardware such as the NVIDIA Jetson Orin.

9 Planned Future Work: Weeks 5–8

The work completed in Weeks 1 through 4 establishes a robust foundation. The remaining four weeks of the project will extend the framework in three directions, each targeting one of the identified research gaps.

9.1 Week 5: Semantic Flow Loss for Temporal Consistency

The most immediate limitation of DS-WGC v2 is that it operates on individual frames independently. In a video sequence, independent per-frame processing causes the enhanced luminance to fluctuate between frames, producing a flickering artefact that is highly disruptive to both human perception and temporal object tracking algorithms.

The planned Semantic Flow Loss addresses this by computing optical flow estimates between consecutive frame pairs using a lightweight off-the-shelf estimator (PWC-Net or RAFT-small) and enforcing that the enhanced high-frequency wavelet coefficients (LH , HL) of identified critical objects remain consistent across the warped frame pair:

$$\mathcal{L}_{SFL} = \sum_{o \in \mathcal{O}} \|LH_e^t(o) - \mathcal{W}(LH_e^{t-1}(o), \mathbf{f}_{t-1 \rightarrow t})\|_1 \quad (24)$$

where $\mathcal{O}$ is the set of critical object bounding boxes detected by a frozen YOLOv8 model, $\mathcal{W}(\cdot, \mathbf{f})$ is the backward-warp operator using flow field $\mathbf{f}$, and $LH_e^t(o)$ denotes the enhanced horizontal-edge sub-band cropped to object o in frame t . This loss is zero-reference — it uses the network’s own outputs across time as a self-consistency signal.

9.2 Week 6: Foundation Model Semantic Distillation

Vision foundation models such as CLIP, DINO, and DINOv2 encode rich semantic representations that correlate strongly with object detection performance. However, their computational cost ($>200\text{M}$ parameters) makes direct inference-time use infeasible for real-time ADAS. Week 6 will explore distilling these representations into the DS-WGC training loop as a static perceptual loss:

$$\mathcal{L}_{distill} = \|\mathcal{F}_{DINO}(I_{enh}) - \mathcal{F}_{DINO}(I_{gt})\|_2 \quad (25)$$

where $\mathcal{F}_{DINO}$ extracts features from a frozen DINOv2 ViT-S backbone at a single inference step per training batch. At test time, the foundation model is not used; its knowledge is embedded in the enhanced student weights through the training-time gradient signal. This

addresses Gap 2 identified in the literature review without incurring any inference-time overhead.

9.3 Week 7: Dark Channel Prior Integration and Physics-Informed Loss

The Dark Channel Prior (DCP), originally developed for single-image dehazing, provides a computationally inexpensive estimate of local minimum pixel values across colour channels. In a haze-free image, dark channels are expected to be near zero; deviations indicate either haze or incorrect colour balance. For low-light enhancement, the DCP can serve as a spatial prior to identify regions that are insufficiently bright (high dark channel values relative to the local mean) versus regions that are saturated:

$$DC(x) = \min_{y \in \Omega(x)} \min_{c \in \{R, G, B\}} I_c(y) \quad (26)$$

Integrating this prior as an additional loss term provides physics-grounded localisation of shadow and glare regions without requiring any labelled data, strengthening the self-supervised signal.

9.4 Week 8: Final Integration, Benchmark Evaluation, and Embedded Deployment Study

The final week will integrate all components, conduct a thorough comparative evaluation against published state-of-the-art methods on the LOL, DICM, NPE, and ExDark benchmarks, and perform a deployment feasibility study. This will include profiling the model on NVIDIA TensorRT with FP16 quantisation to estimate real-time throughput on Jetson Orin hardware, and documenting the accuracy-latency tradeoff.

A final ablation study will systematically remove each component — semantic flow loss, foundation distillation, DCP prior, glare suppression, wavelet decomposition — to quantify the marginal contribution of each innovation to the final PSNR, SSIM, and mAP on ExDark.

10 Discussion

10.1 Why Self-Supervision Matters for ADAS

The practical motivation for the self-supervised approach deserves emphasis. In controlled laboratory settings, paired datasets like LOL provide sufficient diversity for training a supervised model. On a production vehicle, however, the distribution of nighttime scenes is effectively unbounded: varying weather conditions, geographically specific streetlamp spectra (high-pressure sodium vs. LED), the unique optical signature of the vehicle’s own camera and

lens assembly, and the evolving patterns of urban illumination infrastructure all contribute to a distribution shift that no static training set can fully capture. A self-supervised model, trained on the images it actually encounters in deployment, has a structural advantage in adapting to this variability.

10.2 The PSNR–Perception Gap

The downstream YOLOv8 experiment (Section 8.3) illustrates a well-known but often under-appreciated phenomenon: pixel-level reconstruction quality and semantic perceptual quality are not the same thing. DS-WGC v2’s +27% PSNR improvement over Zero-DCE corresponds to a +67% detection improvement. This super-linear relationship suggests that the wavelet-domain denoising and glare suppression are specifically recovering the high-frequency edge features that object detectors weight most heavily, rather than uniformly improving pixel accuracy. This finding supports the design decision to process the LH/HL sub-bands with edge-preserving gates rather than aggressive enhancement, since preserving these features intact appears to be more valuable than attempting to “improve” them.

10.3 Limitations

The current framework has several limitations that future work will address. First, the evaluation is conducted on a small test set (15 pairs) and results should be interpreted with appropriate statistical caution. Second, the synthetic degradation model used as a dataset fallback does not faithfully reproduce all real-world low-light artefacts, including lens flare, chromatic aberration, and rolling shutter effects from fast-moving sensors. Third, the temporal consistency component remains unimplemented and the current model will produce flickering in video sequences. Finally, the glare mask is learned implicitly and is not explicitly supervised, which may lead to inconsistent behaviour on out-of-distribution lighting scenarios.

11 Conclusion

This report documents the first four weeks of research into the DS-WGC (Dual-Scale Wavelet Glare-Consistency) framework, a novel self-supervised architecture for low-light image enhancement targeted at ADAS applications. Starting from the Zero-DCE baseline, each weekly development milestone has introduced a principled architectural innovation grounded in signal-processing theory: Haar wavelet decomposition for frequency-domain separation, sub-band-specific processing networks for illumination, denoising, and edge preservation, and inversion-guided glare suppression for high-dynamic-range scene handling.

The experimental results demonstrate consistent, progressive improvement across all three

evaluation dimensions: pixel-level PSNR improves by +13.49 dB over the unprocessed input, SSIM improves by +0.4550, and YOLOv8 average detections increase by +3.48 objects per image. These improvements are achieved entirely without paired training data, confirming the viability of the self-supervised, signal-processing-assisted paradigm for this problem.

The planned continuation through Weeks 5 to 8 will address the remaining challenges of temporal consistency, semantic alignment with foundation model knowledge, and physics-informed spatial priors, with the goal of producing a production-grade enhancement module that can serve as a drop-in front-end for any ADAS perception stack operating in low-light conditions.

References

- [1] C. Guo, C. Li, J. Guo, C. C. Loy, J. Hou, S. Feng, and R. Chen, “Zero-Reference Deep Curve Estimation for Low-Light Image Enhancement,” *Proc. IEEE/CVF CVPR*, pp. 1780–1789, 2020.
- [2] UMFNet: Frequency-Guided Multi-Scale Fusion with Dynamic Noise Suppression for Robust Low-Light Object Detection, *MDPI Applied Sciences*, 2025.
- [3] VELIE: A Vehicle-Based Efficient Low-Light Image Enhancement Method for Intelligent Vehicles, *MDPI Sensors*, vol. 24, no. 4, 2024.
- [4] WTEFNet: Real-Time Low-Light Object Detection for Advanced Driver Assistance Systems, arXiv:2505.23201, 2025.
- [5] Temporal Consistency-Aware Zero-Shot Illumination-Guided Low-Light Video Enhancement, *IEEE Xplore*, 2024.
- [6] C. Wei, W. Wang, W. Yang, and J. Liu, “Deep Retinex Decomposition for Low-Light Enhancement,” *Proc. British Machine Vision Conference (BMVC)*, 2018.
- [7] Rethinking Zero-DCE for Low-Light Image Enhancement, *ResearchGate*, 2024.
- [8] Real-time Automotive Night-vision System for Glare Suppression, *ResearchGate*, 2021.
- [9] V. Lore, A. Akintayo, and S. Sarkar, “LLNet: A Deep Autoencoder Approach to Natural Low-Light Image Enhancement,” *Pattern Recognition*, vol. 61, pp. 650–662, 2017.
- [10] Improving Visual and Downstream Performance of Low-Light Enhancer with Vision Foundation Models, *CVPR 2025*, CVF Open Access.
- [11] Retinex-MEF: Retinex-based Glare Effects Aware Unsupervised Multi-Exposure Image Fusion, *ICCV 2025*, CVF Open Access.
- [12] Deep Learning Based Low Light Enhancements for Advanced Driver-Assistance Systems at Night, *ResearchGate / IEEE*, 2024.
- [13] Perception and Sensing for Autonomous Vehicles under Adverse Weather Conditions: A Survey, *ResearchGate*, 2023.
- [14] Hybrid Deep Learning Approach for Low-Light Image Enhancement Based on Attention-Guided Residual Networks, *IJETA*, 2025.
- [15] G. Jocher, A. Chaurasia, and J. Qiu, “YOLOv8 by Ultralytics,” <https://github.com/ultralytics/ultralytics>, 2023.

- [16] A. Coates, A. Ng, and H. Lee, “An Analysis of Single-Layer Networks in Unsupervised Feature Learning,” *Proc. AISTATS*, 2011.
- [17] Unsupervised Low-Light Video Enhancement with Spatial-Temporal Decomposition, *IJCAI 2025*.
- [18] M. Caron et al., “Emerging Properties in Self-Supervised Vision Transformers,” *Proc. ICCV 2021*.

A Code Architecture Overview

The project codebase is structured as a single, self-contained Python script designed for Google Colab execution. The key components are:

Section 0–2: Environment Setup. Package installation, device configuration, seed initialisation, and dataset loading (with three-tier fallback).

Section 3–4: Shared Utilities. Metric computation functions (`calc_psnr`, `calc_ssim`), tensor-to-NumPy conversion, and the unified `LOLDS` PyTorch dataset class with augmentation support.

Section 5: Loss Primitives. Reusable loss modules (`SpaL`, `ExpL`, `CoLL`, `TVL`) shared across all model variants.

Section 6: Zero-DCE. The baseline network, its loss, training loop, and evaluation.

Section 7: DS-WaveletNet. Haar DWT/IDWT modules, sub-band processing branches, wavelet loss, training loop.

Section 8: DS-WGC v2. GlareDetector, GlareSuppress, DSWGCv2 full network, GlareLoss, training loop.

Section 9: Figures. Automated generation of all comparison figures and sub-band visualisations.

Section 10: YOLOv8 Evaluation. Downstream detection counting and bar plot generation.

Section 11: Summary Table. Console output of all comparative metrics.

B Haar Wavelet Mathematical Properties

The Haar wavelet is the simplest member of the orthogonal wavelet family. Its filter bank consists of a low-pass filter $h = \frac{1}{\sqrt{2}}[1, 1]$ and a high-pass filter $g = \frac{1}{\sqrt{2}}[1, -1]$. For a 2D signal,

the four sub-bands correspond to the four possible combinations of applying h or g in each spatial dimension.

The perfect reconstruction property guarantees:

$$\text{IDWT}(\text{DWT}(\mathbf{x})) = \mathbf{x} \quad (27)$$

for any input $\mathbf{x}$ of even spatial dimensions. This property is essential for the DS-WGC framework because it means that any enhancement applied in the wavelet domain is faithfully transferred back to the spatial domain without residual reconstruction error.

The energy compaction property of wavelets means that for natural images (which are statistically smooth), most of the signal energy is concentrated in the LL sub-band. In a low-light image with additive white noise, the noise energy is approximately equally distributed across all four sub-bands, but the HH band contains proportionally more noise relative to signal, making it the optimal target for thresholding-based denoising.

C Project Timeline

Table 5: DS-WGC project timeline across 8 weeks

Week	Focus	Key Deliverable	Status
1	Dataset & Setup	LOL pipeline, EDA figures	Complete
2	Zero-DCE Baseline	Trained model, PSNR/SSIM	Complete
3	DS-WaveletNet	Haar DWT/IDWT, sub-band analysis	Complete
4	Glare Suppression (DS-WGC v2)	GlareDetector, YOLOv8 eval	Complete
5	Semantic Flow Loss	Temporal consistency for video	<i>In Progress</i>
6	Foundation Distillation	DINO perceptual loss	<i>Planned</i>
7	DCP Prior Integration	Physics-informed shadow/glare	<i>Planned</i>
8	Final Benchmark & Deployment	TensorRT, ExDark, full ablation	<i>Planned</i>